

Air University



Organization & Assembly (Lab)

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Lab Task # 08

Program 1:

Display the Entire IBM Character Set (256 Characters)

Code

```
01 .model small
02 .stack 100h
03 .code
04
05 main proc
06     ; Initialize counter to 256
07     mov cx, 256
08
09     ; Start with character code 0 in BL
10     mov bl, 0
11
12 print_loop:
13     ; Load current character into DL
14     mov dl, bl
15
16     ; Print the current character
17     mov ah, 2
18     int 21h
19
20     ; Print a space after the character
21     mov dl, 32 ; ASCII code for space
22     mov ah, 2
23     int 21h
24
25     ; Increment character code in BL
26     inc bl
27
28     ; Decrement counter
29     dec cx
30
31     ; Jump back if counter is not zero
32     jnz print_loop
33
34     ; Terminate the program
35     mov ah, 4ch
36     int 21h
37
38 main endp
39 end main
```

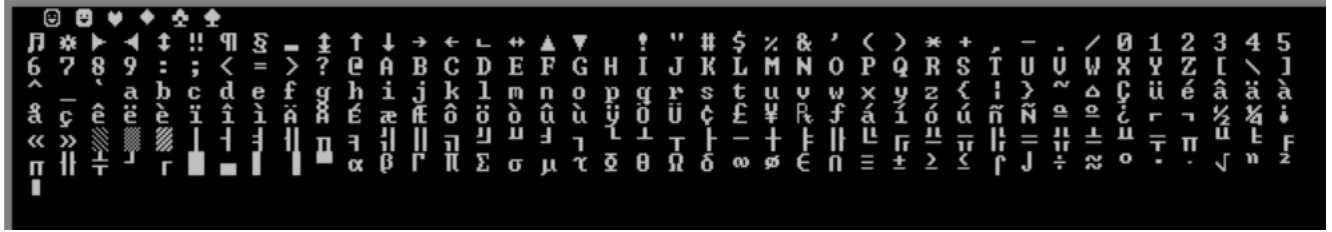
Explanation:

- **Initialization:** The loop counter CX is set to 256, and the character code register BL starts at 0.
- **Loop Body:** In print_loop, the current code from BL is printed using the **DOS Character Output function** (INT 21h, AH=2), followed by a space (ASCII 32).
- **Iteration:** BL is incremented to get the next character code, and CX is decremented.
- **Loop Control:** The JNZ (Jump if Not Zero) instruction repeats the loop until CX is 0.

- **Termination:** The program exits using the **DOS Terminate Program function** (INT 21h, AH=4Ch).

Output

 emulator screen (159x51 chars)



Program 2:

Print Uppercase English Letters (A to Z)

Code

```
.model small
.stack 100h
.code

main proc
    mov cx, 26          ; Initialize counter to 26 (for A to Z)
    mov bl, 'A'         ; Start with 'A' in BL (BL holds the current letter code)

loop_start:
    mov dl, bl          ; Load current letter into DL
    mov ah, 2           ; DOS interrupt 21h, function 2 (write to stdout)
    int 21h             ; Print the current character

    ; Print a space after the character
    mov dl, 32          ; ASCII code for space
    mov ah, 2
    int 21h

    inc bl              ; Increment letter code in BL
    dec cx              ; Decrement counter
    jnz loop_start      ; Jump back if counter is not zero

    ; Terminate the program (AH=4Ch for INT 21H)
    mov ah, 4ch
    int 21h

main endp
end main
```

Explanation:

This assembly program prints the 26 uppercase letters A through Z separated by a space, using a loop.

- **Initialization:** The counter CX is set to 6 (the number of letters), and the starting letter 'A' is loaded into BL.
- **Loop:** The code iteratively prints the character in BL using the **DOS Character Output function** (INT 21h, AH=2).
- **Spacing:** A space (ASCII 32) is printed immediately after each letter for separation.
- **Control:** BL is incremented to get the next letter (A→B→C...), CX is decremented, and the loop repeats using JNZ until all 26 iterations are complete.
- **Exit:** The program terminates cleanly with the **DOS Exit function** (INT 21h, AH=4Ch).

Output

```
D:\>D:\test
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
D:\>[_]
```

THE END
